# Echo AI — Functional Product Design

**Status:** Product baseline
**Prepared:** 2026-11-11
**Design basis:** Current product documentation and the user's Echo feature history.
**Audience:** Product owner, designers, implementers, and testers.

## 1. Product intent

Echo is a desktop-first AI assistant that brings conversation, voice, personal productivity, and carefully gated computer automation into one interface. It should help a person get work done without hiding when it is using AI, touching local data, operating a device, or communicating with an external service.

The primary product experience is a local web interface on the computer running Echo. The user can speak or type naturally, then use focused workspaces for camera controls, system care, web automation, research, and personal planning. Capabilities that can affect privacy, the operating system, a microphone, a purchase, or another person's data require a clear user action and an observable status.

### Product principles

1. **User-directed:** background monitoring and consequential actions are off until the user explicitly enables them.
2. **Understandable:** Echo reports what it recognized, what it did, and when it could not determine an outcome.
3. **Fail safe:** uncertain recognition, missing permissions, lock state, incomplete data, or network errors must not be reported as success.
4. **Local by default for device features:** camera, system, and device monitoring do not send their raw data to an AI service.
5. **Explicit authority:** a user grants the permission needed for each consequential workflow; permission for one feature does not imply permission for another.
6. **Reversible where possible:** users can stop, pause, disarm, undo a configuration, or restore a saved system setting. Irreversible actions are separately confirmed and their outcome is made clear.

## 2. Scope baseline

- **Discovery method:** reviewed the current README feature descriptions and recent product conversation; no project constitution, clarification document, architecture summary, or knowledge graph was present.
- **Functional domains identified:** 9.
- **Domains in scope:** all 9, as the current Echo product baseline: assistant and device commands; voice and wake; dashboard and settings; workspace and planning; presence and camera assistance; system care; web automation and website scanning; personal exposure checks; business knowledge and tenant administration.
- **Assumption:** this document describes the existing product direction and documented behavior. It is not a request to enable every feature by default or to expand the system's permissions.

# 3. Users and access modes

## User types

- **Personal user:** uses Echo on their own computer and manages their own settings, notes, devices, and automation.
- **Demo visitor:** uses a deliberately restricted demonstration of Echo.
- **Enterprise platform administrator:** creates and manages enterprise tenants and their initial administrators.
- **Enterprise administrator:** manages users and approved documents within their own tenant.
- **Enterprise member:** asks questions of their tenant's approved knowledge base and provides feedback.

## Operating modes

| Mode | Intended use | Functional boundary |
|---|---|---|
| Personal | User's own trusted computer | Local features are available when the user explicitly enables them. It is not an authenticated multi-user service. |
| Demo | Guided demonstration | Requires configured access; computer-control and monitoring features are disabled. Demo knowledge is not confidential or tenant-isolated. |
| Enterprise | Local multi-tenant trial | Account and tenant boundaries apply. Tenant members access only approved tenant documents and their own permitted functions. Device-control features remain unavailable. |

The interface must show the current mode and must not imply that demo or enterprise users have personal-device authority.

# 4. Information architecture

1. **Dashboard:** entry point, feature cards, status, and action filtering.
2. **Assistant:** text and voice conversation, command results, and explicit confirmations.
3. **Workspace:** speech notes, reading list, translation, and summarization.
4. **Planning and memory:** tasks, timers, reminders, agenda, and user-directed memory management.
5. **Presence & Security:** opt-in away detection and Windows screen locking.
6. **Posture & Gestures:** camera consent, calibration, hand scan, practice, action arming, live recognition, and stop controls.
7. **System Care:** local network observations, event/log analysis, and confirmed hardware-profile actions.
8. **Web Automation:** stock watchers, browser workflows, selector health, and activity history.
9. **Data Leak Check:** authorized personal identifiers, results, and rescan settings.
10. **Website Scanner:** website inspection and scan results.
11. **Settings:** language, voice, AI provider, profile, and feature preferences.
12. **Enterprise knowledge and administration:** source-grounded questions, document management, user management, and tenant-scoped activity.

Each feature page should include a concise explanation, prerequisites, permission/consent controls, current status, primary action, stop or recovery action, and a user-readable outcome.

# 5. Primary user journeys

### Journey A — Ask Echo for help (Priority: P1)

The user opens the dashboard, types or speaks a request, reviews Echo's answer or proposed action, and sees the result. For consequential local actions, the user reviews a clear confirmation before execution.

**Independent test:** complete one information question and one permitted local command from the assistant interface.

**Acceptance scenarios:**

1. **Given** a configured AI provider, **when** the user asks a question, **then** Echo displays a response and identifies unavailable or failed services instead of inventing a successful result.
2. **Given** a command that changes or deletes local data, **when** the user submits it, **then** Echo presents the target and asks for confirmation before performing the change.
3. **Given** no configured AI provider, **when** the user requests an AI-dependent task, **then** Echo explains the missing setup and does not present a fabricated answer.

### Journey B — Capture, organize, and revisit work (Priority: P1)

The user creates a transcript-based note, saves an article for later, translates or summarizes permitted content, and uses tasks, timers, or reminders to organize follow-up work.

**Independent test:** create a note, add and find a reading-list entry, and set a reminder.

**Acceptance scenarios:**

1. **Given** supported text or a supported document, **when** the user requests translation or a summary, **then** Echo presents editable output and identifies material it could not read.
2. **Given** a public article URL, **when** the user saves it, **then** Echo stores its URL and readable metadata and allows the user to find, mark, or remove it.
3. **Given** a timer or reminder request, **when** the user saves it, **then** Echo shows its scheduled state and later surfaces the notification.

### Journey C — Enable safe camera assistance (Priority: P1)

The user grants camera consent, starts the assistant, calibrates posture if wanted, and performs a one-hand gesture scan. Recognition is practice-only until the user explicitly arms actions. The user can disarm or stop at any time.

**Independent test:** with synthetic camera landmarks, scan, arm, recognize, disarm, and stop without operating a physical device.

**Acceptance scenarios:**

1. **Given** the user has not consented or has not completed a scan, **when** the camera page opens, **then** computer actions remain off.
2. **Given** an armed session and a clear pinch, **when** the user holds the pinch briefly and changes fingertip spacing, **then** Echo adjusts system volume in small steps and reports the action.

3. **Given** ambiguous or lost hand tracking, **when** an action might otherwise be triggered, **then** Echo does nothing and does not catch up missed volume changes later.
4. **Given** the user disarms, stops, or the desktop is locked, **when** a gesture is seen, **then** it causes no computer action.

### Journey D — Protect a workstation when away (Priority: P2)

The user selects a camera and delay, grants permission, and explicitly starts presence detection. Echo locks Windows after no face is detected for the configured interval and may provide a return notification. The user unlocks through Windows, not through Echo.

**Independent test:** simulate absent/present observations and verify locking, return notification, stop, and camera-error behavior.

**Acceptance scenarios:**

1. **Given** active detection and no face for the configured interval, **when** the interval expires, **then** Echo requests Windows screen lock.
2. **Given** the user returns and remains visible, **when** return detection completes, **then** Echo may notify them but does not unlock Windows.
3. **Given** camera or lock failure, **when** the failure occurs, **then** Echo reports it and stops detection rather than claiming protection is active.

### Journey E — Review local system signals (Priority: P2)

The user selects local LAN and/or supported system-log checks, configures intervals, grants permission, and starts monitoring. The user reviews unknown devices or event summaries and decides what to trust or investigate. Echo does not block devices or raise its own privileges.

**Independent test:** feed known and unknown synthetic devices/events and verify that only the configured checks run and notifications avoid exposing private detail in unrelated dashboard surfaces.

### Journey F — Monitor a product or perform an authorized transaction (Priority: P2)

The user configures a product watcher, chooses notification, cart, manual checkout, or an explicitly authorized order/reservation workflow, and reviews the activity. Any binding transaction requires a separate, one-time confirmation with product, quantity, total budget, and currency. Uncertain submission outcomes are treated as unknown, not as success or as permission to retry.

**Independent test:** simulate in-stock/out-of-stock transitions, price changes, selector failures, and final confirmation outcomes without submitting a real transaction.

### Journey G — Check the user's own public exposure (Priority: P2)

The user identifies an email, username, or domain as theirs, selects a check, and reviews results and limitations. Scheduled checks are opt-in. Echo does not claim that a username match proves identity or compromise.

**Independent test:** verify ownership confirmation, result-source labels, scheduled opt-in, and secret clearing behavior using a mock provider.

### Journey H — Answer from approved business knowledge (Priority: P3)

An enterprise member asks a question, Echo retrieves approved tenant documents, and the answer includes source references. An administrator manages only that tenant's users and documents; a platform administrator manages tenant setup.

**Independent test:** verify an answer cites its source and that a user from one tenant cannot view another tenant's documents or administrative activity.

# 6. Functional requirements

### Assistant and interaction

- **REQ-001:** Echo MUST accept text input and, when configured and permitted, voice input; it MUST clearly distinguish a completed action from a proposed or failed action.
- **REQ-002:** Echo MUST route supported local commands through a bounded, user-visible command flow and MUST request confirmation before destructive file changes, submission, or other consequential actions.
- **REQ-003:** Echo MUST expose dashboard navigation, current feature status, and search/filtering without starting a camera, monitor, or transaction when the user merely opens a page.
- **REQ-004:** Echo MUST support configurable speech input/output and report when a requested speech or AI service is unavailable. Wake listening MUST be separately startable and MUST NOT silently authorize computer-control actions.

### Workspace and planning

- **REQ-005:** Echo MUST allow the user to create and manage notes, reading-list entries, tasks, timers, and reminders, and to search or retrieve those items through the relevant interface.
- **REQ-006:** Echo MUST allow supported text, documents, and public URLs to be summarized or translated, communicate format/size/reading limitations, and leave generated content reviewable by the user.
- **REQ-007:** Echo MUST allow the user to inspect, add, recall, and clear persistent personal memory; memory MUST NOT be implied when a fact has not been saved.

### Presence and camera assistance

- **REQ-008:** Echo MUST obtain explicit permission and show a distinct running status before capturing camera frames for presence, posture, or gestures.
- **REQ-009:** Presence detection MUST lock the operating-system session only after its configured absence condition; return detection MUST only notify and MUST NOT authenticate or unlock the user.
- **REQ-010:** Posture assistance MUST let the user calibrate a comfortable baseline, configure a reminder delay, and stop reminders; its messages MUST not present webcam posture estimates as medical advice.
- **REQ-011:** Gesture actions MUST remain disarmed until the user completes a fresh, explicit hand scan for the current detection session and chooses to enable actions.
- **REQ-012:** With extra gestures enabled, a held pinch with three other fingers extended MUST provide bounded incremental system-volume control; greater normalized fingertip spacing raises volume and smaller spacing lowers it. A stable pinch MUST NOT repeat volume changes solely because it is held.
- **REQ-013:** Gesture controls MUST preserve the distinction between an open hand swipe, a stationary pointing gesture for play/pause, pinch volume, single-hand gestures, and two-hand gestures. Unclear, overlapping, or lost tracking MUST cause no action.
- **REQ-014:** Gesture actions MUST be disabled in practice mode, after disarming, on lock/unavailable desktop, and after an action error until the user explicitly reactivates them. A microphone mute MUST NOT silently unmute.

### System care

- **REQ-015:** LAN and system-log monitoring MUST be individually configurable, explicitly started and stopped, and report coverage gaps, permission errors, and uncertain observations.
- **REQ-016:** Unknown LAN devices MUST be presented as observations for user review; Echo MUST NOT automatically block them or represent a device observation as proof of intrusion.
- **REQ-017:** Any process or hardware-profile action MUST show the affected targets and require confirmation. Protected processes and other users' sessions MUST remain out of scope, and Echo MUST provide an available recovery path for reversible system-setting changes.

### Web automation, scanning, and exposure checks

- **REQ-018:** Product watchers MUST allow the user to test without action, pause, resume, edit, or remove a watcher; ordinary cart and manual-checkout modes MUST NOT submit a final order or reservation.
- **REQ-019:** An order or reservation MUST require distinct workflow and one-time transaction authorization, a configured total budget, exact transaction checks, and a visible confirmation outcome. Missing or changed product, quantity, price, destination, or success evidence MUST block or leave the outcome explicitly unknown. Echo MUST NOT automatically retry an uncertain submission.
- **REQ-020:** Automatic selector repair MUST be limited to non-transactional elements with a sufficiently clear match. Final payment, order, and reservation controls MUST require explicit user-configured selectors and renewed authorization when they change.
- **REQ-021:** Website scans and personal-exposure checks MUST show their scope, data sources, limitations, and failures. Exposure checks MUST require the user to confirm they own or may check the supplied identifiers.

### Business access and privacy

- **REQ-022:** Business-knowledge answers MUST cite the approved source documents used; lack of a relevant source MUST be reported rather than replaced with an unsupported answer.
- **REQ-023:** Enterprise users, documents, and activity MUST be isolated by tenant and role. Demo mode MUST be clearly labeled and MUST disable personal-device control.
- **REQ-024:** Echo MUST explain which data stays local and which content is sent to an external AI or service before use where that distinction affects user privacy. It MUST provide stop controls for continuous monitoring and avoid success-shaped fallbacks when status is unknown.

## 7. Edge cases and recovery

- Camera permission is denied, the camera is occupied, or another feature is using the same camera: show the conflict and do not claim detection is active.
- The computer is locked, asleep, or unavailable: suppress gesture actions and require explicit reactivation when the policy requires it.
- Gesture recognition is interrupted or hands overlap: reset the gesture's movement baseline; do not perform a delayed action.
- A selected microphone disappears: require the user to select another device; do not silently substitute one.
- A log source is unreadable or a network device cannot be probed: disclose the missing coverage and do not infer "clean" or "safe."
- A process has changed identity between preview and execution: refuse to terminate it and require a new preview.

- A product page changes, price exceeds budget, a CAPTCHA appears, or the checkout result cannot be confirmed: stop, report the reason/unknown state, and never retry a transaction automatically.
- An external provider is unavailable or rate-limits a request: show the provider error and preserve the user's ability to retry manually where safe.
- A user crosses demo/enterprise/personal boundaries: retain each mode's access restrictions and do not expose another mode's data or controls.

## 8. Key entities

- **User and access mode:** person, role, tenant, and current operating mode.
- **Conversation and command:** user request, assistant response, proposed action, confirmation, and outcome.
- **Preference and permission:** user-selected settings, consent, and feature enablement state.
- **Workspace item:** note, reading-list entry, task, timer, reminder, or saved memory.
- **Monitoring session:** feature, target device/source, start/stop state, intervals, consent, and latest status.
- **Gesture scan and action:** current-session scan, practice/armed state, recognized gesture, action, and result.
- **System observation and policy:** device/log event, trust decision, confirmed resource policy, preview, and restoration information.
- **Web watcher and transaction attempt:** target, monitoring policy, action class, budget, authorization, attempt state, and confirmation evidence.
- **Exposure check:** user-authorized identifier, provider, result, and previous check state.
- **Knowledge source and answer:** tenant-scoped approved document, retrieved evidence, answer, and feedback.

## 9. Non-functional expectations

- The interface MUST work on a desktop browser at common desktop and mobile viewport widths without hiding primary stop, consent, or status controls.
- Every persistent monitor MUST have an explicit running/stopped state and a reachable stop action. Restarting Echo MUST NOT silently restart sensitive monitoring or transaction authorization.
- Status, error, and action feedback MUST be understandable in the selected language and MUST distinguish "off," "running," "paused," "unknown," and "failed."
- Camera and local system data MUST remain local unless the user initiates a specifically explained external service request. Secrets MUST NOT be exposed in the browser or written into unprotected project files.
- The design MUST support keyboard and screen-reader access to primary actions, labels, consent, progress, status, and error messages.

## 10. Success criteria

- **SC-001:** A first-time user can find the assistant, submit a basic request, and understand its result without assistance in under 3 minutes.
- **SC-002:** In usability testing, at least 90% of participants can tell whether camera monitoring is off, running, or uncertain before leaving the camera page.

- **SC-003:** In a controlled set of ambiguous, interrupted, overlapping, and disarmed gesture cases, zero computer actions are executed.
- **SC-004:** At least 90% of test users can explain how to disarm/stop camera actions and how to resume them after a lock or interruption.
- **SC-005:** Every transaction test with a mismatched product, quantity, budget, currency, final control, or missing confirmation is blocked or reported as unknown; none is reported as confirmed.
- **SC-006:** Every enterprise access-control test prevents cross-tenant document and activity access.
- **SC-007:** For each monitoring page, at least 90% of test users can locate its stop control and identify whether it continues after the page is closed.

## 11. Assumptions and boundaries

- This design treats Echo as a Windows-first desktop product with a local web interface; selected workspace and document functions may also run on other supported operating systems.
- Echo does not unlock Windows with face recognition. Windows Hello or a normal Windows credential handles authentication.
- A detected LAN device or log pattern is a signal for the user to review, not proof of an attack or an automatic enforcement decision.
- Camera gestures control the operating-system media/volume interfaces; they do not establish an authenticated Spotify account or guarantee that Spotify is the active media session.
- Safe product monitoring, cart actions, and final transactions are separate user-selected levels of authority. A watcher is not permission to purchase.
- This document is a functional baseline, not a claim of security certification, medical accuracy, guaranteed detection, or guaranteed compatibility with every external site or device.

## 12. Out of scope

- Unlocking the computer by ordinary webcam face recognition.
- Automatically blocking unknown LAN devices or declaring an intrusion based only on a device/log observation.
- Bypassing CAPTCHAs, bot protections, authentication, payment authorization, or website access controls.
- Unbounded autonomous purchases, retries after uncertain submission, or purchases without a separate, explicit transaction authorization.
- Clinical posture assessment or biometric identity profiling.
- Direct fan-speed control on hardware that does not expose a supported control; system cooling policy is not a guaranteed RPM setting.
- Treating AI output as authoritative without user review or source evidence.